exp18.py - C:\Users\pings\Desktop\exp18.py (3.14.2)

File Edit Format Run Options Window Help

```
import numpy as np
import pandas as pd

df = pd.read_csv(r'C:\Users\pings\Desktop\iris - iris.csv')
# make column / class names look like the manual
df.columns = ['sepal.length', 'sepal.width', 'petal.length', 'petal.width', 'variety']
df['variety'] = df['variety'].str.capitalize()
df.info()

print(df.variety.value_counts())
print(df.head())

features = df.iloc[:, :-1].values
label = df.iloc[:, 4].to_numpy()

from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier

xtrain, xtest, ytrain, ytest = train_test_split(features, label, test_size=.2, random_state=1)
model_KNN = KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain, ytrain)

print(model_KNN.score(xtrain, ytrain))
print(model_KNN.score(xtest, ytest))

from sklearn.metrics import confusion_matrix
print(confusion_matrix(label, model_KNN.predict(features)))

from sklearn.metrics import classification_report
print(classification_report(label, model_KNN.predict(features)))
```

Ln: 30 Col: 64

IDLE Shell 3.14.2

File Edit Shell Debug Options Window Help

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:\Users\pings\Desktop\exp18.py ==========
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    str
dtypes: float64(4), str(1)
memory usage: 7.2 KB
variety
Setosa        50
Versicolor    50
Virginica     50
Name: count, dtype: int64
   sepal.length  sepal.width  petal.length  petal.width variety
0           5.1          3.5           1.4          0.2  Setosa
1           4.9          3.0           1.4          0.2  Setosa
2           4.7          3.2           1.3          0.2  Setosa
3           4.6          3.1           1.5          0.2  Setosa
4           5.0          3.6           1.4          0.2  Setosa
0.9583333333333334
1.0
[[50  0  0]
 [ 0 47  3]
 [ 0  2 48]]
              precision    recall  f1-score   support

      Setosa       1.00      1.00      1.00        50
  Versicolor       0.96      0.94      0.95        50
   Virginica       0.94      0.96      0.95        50

    accuracy                           0.97       150
   macro avg       0.97      0.97      0.97       150
weighted avg       0.97      0.97      0.97       150

>>>
```

Ln: 43 Col: 0